

PREPARED FOR

T6N, R4W, Mount Diablo Meridian

COORDINATES 38.3459°N 122.3047°W WGS84	ELEVATION 66 ft NAVD88 · MSL	STATE PLANE CA 2 (0402) · N 1,887,922.81 E 6,474,289.05 (US ft, NAD83) -	FEMA ZONE Zone X 06055C0510FF	DECLINATION 13.0° E WMM 2026
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APN 038050009000	ZONING AP	ACRES 0.481 ac
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BEST GNSS WINDOW Sun, Jul 12 · 2:00 PM – 6:00 PM	PDOP 0.97 · 35 sv	NEAREST CORS P261 · 14.1 mi ✓
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Site file · Google Earth
Download KMZ

RECORDER
Napa County Recorder-Clerk · 1127 First St, Napa, CA 94559

NEAREST CORS
P261 · 14.1 mi

GENERATED
-

ASSESSOR
Napa County Assessor · 1127 First St, Napa, CA 94559

NEAREST BM
JT9010 · 0.2 mi

SYSTEM
TopoScout · info@toposcout.com

SITE & SIGNAL REPORT

COORDINATES	ELEVATION	MAGNETIC DECLINATION	FEMA FLOOD ZONE
38.3459, -122.3047	66 ft	13.0° E	X
WGS84 / NAD83	Above MSL (approx)	Compass reads 13.0° too far W	AREA OF MINIMAL FLOOD HAZARD · Panel 06055C0510FF

IONOSPHERIC CONDITIONS

PEAK KP INDEX

4.7

AVG KP

2.6

SOLAR FLUX F10.7

138 sfu

TEC (VTEC) ⚠ DEGRADED

20.7 TECU

-4.3 TECU vs median

ASSESSMENT

MODERATE ACTIVITY

Peak Kp 4.7 (avg 2.6). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

GNSS / DOP SUMMARY

DATE	AVG PDOP	BEST 4-HR WINDOW	AVG SATS	RATING
Sun, Jul 12	0.97	2:00 PM – 6:00 PM	35	Ideal

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location – orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates. For site-specific weather, terrain mask angle, and nearest CORS station, order an On-Demand Site Report.

🔥 WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 12, 2026) · Conditions may change – verify before fieldwork

✖ GPS CONSTELLATION HEALTH

⚠ 1 Satellite Currently Unhealthy – PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13° Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.9–0.9 across survey window
PDOP remains ≤ 0.9 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

Point	Cover Type	Obstruction	Multipath
Center	Dwarf Scrub	Low	Low
North	Dwarf Scrub	Low	Low
South	Dwarf Scrub	Low	Low
East	Dwarf Scrub	Low	Low
West	Dwarf Scrub	Low	Low

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

Sunday, Jul 12

WEEKEND

Overcast

10 mph

GO

High 90°F

Low 56°F

Wind max 10 mph

Gusts 12 mph

Precip 0.00"

Sunrise 5:56 AM

Sunset 8:33 PM

Waning Crescent 6%

GNSS: Avg PDOP 0.97 · Best 4-hr window 2:00 PM – 6:00 PM (PDOP 0.93, 35 sats avg)

GNSS / DOP - HOURLY

	PDOP	SATS	RATING
6:00 AM	0.87	36 sv	IDEAL
7:00 AM	0.91	34 sv	IDEAL
8:00 AM	1.06	29 sv	GOOD
9:00 AM	1.00	33 sv	GOOD
10:00 AM	1.00	29 sv	GOOD
11:00 AM	1.06	27 sv	GOOD
12:00 PM	0.96	29 sv	IDEAL
1:00 PM	1.00	28 sv	GOOD
2:00 PM	0.90	34 sv	IDEAL
3:00 PM	0.98	33 sv	IDEAL
4:00 PM	0.98	34 sv	IDEAL
5:00 PM	0.86	38 sv	IDEAL
6:00 PM	1.01	32 sv	GOOD
7:00 PM	1.04	30 sv	GOOD



GO - GOOD CONDITIONS

Partly cloudy | Wind: 10.1 mph (gusts 11.9 mph) | Precip: 0" | Temp: 56.3°–90°F



DRONE WIND ANALYSIS



GO Peak: 10.1/11.9 mph S

* Sunrise: 5:56 AM Sunset: 8:33 PM fly window clipped to daylight

Best flight window: 6:00 AM – 9:00 PM (15 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≠turbulence	DIR
5 AM	3/5 mph		SSE
6 AM	4/7 mph		ESE
7 AM	2/3 mph		ESE
8 AM	3/3 mph		W
9 AM	3/4 mph		SW
10 AM	5/5 mph		SSW
11 AM	6/7 mph		SSE
12 PM	5/6 mph		S
1 PM	6/6 mph		S
2 PM	7/7 mph		S
3 PM	7/7 mph		S
4 PM	10/11 mph		SSW
5 PM	9/11 mph		SSW
6 PM	9/11 mph		SSW
7 PM	8/11 mph		S
8 PM	7/12 mph		SSE

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY
Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107
→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY](#) ↗
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

FIELD PLANNING REPORT

GNSS SURVEY OPERATIONS — NAPA COUNTY, CA

Report Date: Sunday, July 12, 2026 | Survey Start: July 12, 2026

LOCATION: 38.3459, -122.3047 (Napa County) | Analysis Window: 1 day

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VERDICT: GO

PRIMARY RANKING FACTORS: Excellent PDOP geometry (avg 0.97, best window 0.93 with 35 satellites), acceptable weather (overcast, no precipitation, winds 10 mph max), and moderate ionospheric activity (Kp 4.7 peak, TEC 20.7 TECU). All factors support field operations.

BEST SURVEY WINDOW: 2:00 PM – 6:00 PM (Sunday, July 12, 2026)

Rationale: Peak PDOP performance window with 0.93 geometry and 35-satellite constellation. Four-hour afternoon slot avoids morning overcast transition, maximizes satellite visibility, and provides optimal solution quality. Ionospheric delay manageable with extended static sessions given moderate geomagnetic activity.

BEST 4-HOUR STATIC GNSS SESSION: 2:30 PM – 6:30 PM

Deploy dual-frequency receivers minimum; recommend 2-hour minimum occupation per point given Kp 4.7 and TEC estimate quality flag. Ensure receivers auto-exclude PRN 11 (unhealthy). Log raw observables for post-processing ionospheric filtering. Wind conditions (max 10 mph) permit tripod setup without stabilization concerns.

DRONE OPERATIONS ASSESSMENT: APPROVED

Overcast conditions reduce glint risk. Wind max 10 mph is within operational limits for most platforms. No active fire smoke detected in area. Visibility adequate for visual-line-of-sight operations. Recommend flight operations within survey window 2:00 PM – 6:00 PM for optimal ambient light.

FIELD SAFETY WARNINGS

PRN 11 currently unhealthy — no impact on survey geometry (PDOP already accounts for constellation loss), but confirm receiver auto-exclusion before field deployment. Monitor for loss of lock or convergence delays if any legacy single-frequency equipment present. Overcast conditions may suppress visibility on steep terrain — verify site access and egress routes before field work. Peak temperature 90°F — ensure crew hydration and sun protection despite cloud cover.

SITE CONTEXT SUMMARY

Dominant land cover is dwarf scrub with low sky obstruction and low multipath risk; recommend 13° elevation mask. Air quality is good (AQI 45). No active wildfires within range. Satellite maintenance windows noted for PRN 3 (Jul 7–8, Jul 9 onward) and PRN 26 (Jul 17) are outside current survey; informational only.

RECOMMENDATION: Deploy Sunday, July 12, 2:00 PM – 6:00 PM. Conditions support high-precision static and kinematic GNSS operations. Extend static occupations to 2 hours minimum per point to mitigate ionospheric variance.

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COUNTY RECORDING OFFICES

RECORDER / REGISTER OF DEEDS

Napa County Recorder-Clerk
1127 First St, Napa, CA 94559

ASSESSOR / TAX OFFICE

Napa County Assessor
1127 First St, Napa, CA 94559

CORS STATIONS – WITHIN 30 MILES

QR / LINK	STATION ID	NAME / LOCATION	DISTANCE
 NGS ↗	P261	HUNTERHILLCN2004	14.1 mi
 NGS ↗	P198	PETALUMAIRC2004	17.5 mi
 NGS ↗	OHLN	OHLONE PARK	23.5 mi
 NGS ↗	P196	MEACHUMLFLCN2006	24.0 mi
 NGS ↗	CASR	SANTA ROSA CA	24.9 mi
 NGS ↗	P267	DIXONAVIATCN2005	26.3 mi

NGS BENCHMARKS – WITHIN 10 MILES

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
 Datasheet ↗	JT9010	"13"	19.100m / 62.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9018	"13 E"	17.970m / 59.0ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9009	"13"	17.900m / 58.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9017	"13 D"	17.890m / 58.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9034	"20 A"	20.730m / 68.0ft NAVD88	Order	0.3 mi
 Datasheet ↗	JT9012	"13"	19.060m / 62.5ft NAVD88	Order	0.4 mi
 Datasheet ↗	JT9043	"25 A"	18.710m / 61.4ft NAVD88	Order	0.4 mi
 Datasheet ↗	JT9016	"13 D"	18.200m / 59.7ft NAVD88	Order	0.5 mi

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet ↗					

FEMA FLOOD ZONE

X – AREA OF MINIMAL FLOOD HAZARD

Panel: 06055C0510FF

Effective: Sep 29, 2010

[View FEMA Map ↗](#)



DATA COLLECTOR IMPORT – NGS BENCHMARKS

Copy All

NGS BENCHMARK CONTROL POINTS – Copy / Paste to Data Collector

Datum: NAD83 | Vertical: NAVD88 | Source: NGS NDE API

PID	NAME	LAT	LON	ELEV_FT	ORDER
JT9010	13	38.348333	-122.302778	62.7	
JT9018	13 E	38.348333	-122.302778	59.0	
JT9009	13	38.346667	-122.301111	58.7	
JT9017	13 D	38.345000	-122.301111	58.7	
JT9034	20 A	38.345000	-122.309444	68.0	
JT9012	13	38.351667	-122.304444	62.5	
JT9043	25 A	38.340000	-122.302778	61.4	
JT9016	13 D	38.341667	-122.297778	59.7	

USGS STREAM GAUGES – within 25 miles

STATION	DIST	LEVEL	FLOW	FLOOD STATUS
NAPA R NR NAPA CA #11458000	1.5 mi	2.37 ft	4.39 cfs	● Normal
NATHANSON C A SONOMA CA #11458600	9.3 mi	—	—	● Normal
SONOMA C A AGUA CALIENTE CA #11458500	10.4 mi	8.84 ft	2.73 cfs	● Normal
NAPA R NR ST HELENA CA #11456000	14.1 mi	1.44 ft	3.43 cfs	● Normal
SONOMA CREEK A KENWOOD CA #11458433	14.9 mi	3.93 ft	0.19 cfs	● Normal
PUTAH C NR WINTERS CA #11454000	16.8 mi	7.85 ft	578 cfs	— Unknown
SUISUN SLOUGH NR BENICIA CA #11455740	19.6 mi	14.40 ft	-4,650 cfs	— Unknown
POPE C A WALTER SPRINGS CA #11453590	21.1 mi	—	—	— Unknown

USGS LAKES & RESERVOIRS – within 150 miles

STATION	DIST	LEVEL	FULL POOL	LINK
CLEAR LK A LAKEPORT CA #11450000	58.1 mi	5.05 ft	N/R	USGS ↗

CALIFORNIA RESERVOIRS (DWR/CDEC) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Lake Del Valle	60.6 mi	697.01 ft (-9 ft)	706 ft	36K af	68%
Folsom Lake	66.5 mi	448.64 ft (-17 ft)	466 ft	792K af	81%
Lake Oroville	93.4 mi	862.11 ft (-38 ft)	900 ft	2,871K af	81%
New Melones Lake	100.1 mi	1,020.25 ft (-68 ft)	1,088 ft	1,664K af	69%
Black Butte Lake	101.2 mi	—	228 ft	74K af	55%

USBR RESERVOIRS (Bureau of Reclamation) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
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ARMY CORPS RESERVOIRS (USACE) – within 150 miles

RESERVOIR	DIST	POOL ELEV	FULL POOL	STORAGE	% CAP
Del Valle	64 mi	697 ft	695.01 ft	—	100%
Camanche	89 mi	230.3 ft	221.86 ft	—	104%
Englebright Lake-Pool	95 mi	525.1 ft	522.35 ft	—	101%
Farmington Dam-Pool	99 mi	124.9 ft	117.09 ft	—	107%
Black Butte-Pool	101 mi	457 ft	458.36 ft	—	100%
Don Pedro	137 mi	—	N/R	—	—

USGS GROUNDWATER WELLS – within 15 miles

STATION	DIST	WELL DEPTH	WATER LEVEL	DATE	AQUIFER
005N004W32A002M	7 mi	200 ft	87.71 ft BGS	2014-11-05	N100CACSTL
005N004W28R001M	7 mi	51 ft	41.90 ft BGS	1977-10-03	N100CACSTL
004N004W04C001M	8 mi	80 ft	11.20 ft BGS	1977-10-03	N100CACSTL
004N004W05B001M	8 mi	—	54.10 ft BGS	1977-10-03	N100CACSTL
005N004W31R001M	8.1 mi	295 ft	—	—	—
004N004W05D002M	8.3 mi	60 ft	14.60 ft BGS	1977-10-03	N100CACSTL

004N004W06B001M	8.3 mi	201 ft	21.86 ft BGS	2022-06-07	—
004N004W02L001M	8.8 mi	—	9.90 ft BGS	1977-10-02	N100CACSTL

🕒 AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

The Napa Valley Coastal Ranges aquifer (N100CACSTL) consists of fractured volcanic and sedimentary rocks that were deposited and deformed during regional mountain-building over the past 65 million years. Water moves through cracks and pores in these moderately weathered formations, making it a moderately productive aquifer typical of California's Coast Ranges. At this site, water levels range from ~10 to 88 feet below ground surface, indicating a responsive aquifer with variable saturation depending on seasonal recharge and pumping patterns.

FORMATION

Name	Napa Valley Coastal Ranges (undifferentiated Tertiary volcanic and sedimentary rocks)
Age / Rock	Paleocene to Pliocene (65–3 million years ago); fractured andesite, basalt, sandstone, and siltstone.
Extent	800–1200 sq mi (est. USGS/DWR)

DEPTH & RECHARGE

Depth Range	51–295 ft (5–30 floors)
Typical Depth	100–150 ft (10–15 floors)
Recharge Source	Precipitation on hillsides and upland outcrops; lateral throughflow from volcanic highlands
Transit Time	15–40 years to 100 ft depth

WATER QUALITY

pH	est. 7.0–7.6 (est. USGS/DWR)
Hardness	est. 180–350 mg/L CaCO3 (est. USGS/DWR)
TDS	est. 280–520 mg/L (est. USGS/DWR)
Notes	No sample data available — typical for formation; expect moderate hardness and neutral pH characteristic of volcanic/sedimentary Coast Ranges aquifers.

TREND

Trend	Declining long-term trend; water levels in 1977 (9.90–54.10 ft) versus 2022 (21.86 ft) and 2014 (87.71 ft) indicate drawdown consistent with cumulative pumping and variable recharge.
Source	USGS NWIS and California DWR well registry measurements
Seasonal Swing	est. 8–20 ft annual swing (est. USGS/DWR)

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ LOW SUBSIDENCE RISK — No known critically overdrafted basin

Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

🕒 AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU

The Napa Valley area near this coordinate (38.3459, -122.3047) is underlain by the North Coast Ranges aquifer system (N100CACSTL), which benefits from significant groundwater recharge through orographic precipitation and is not classified as a critically overdrafted basin under SGMA. The well-drained soil conditions and moderate groundwater extraction rates in this viticultural region have not induced the sustained water-level declines necessary to cause clay layer compaction and land subsidence. Surveyors working in this area can rely on established benchmarks with confidence that subsidence is not an active geotechnical concern.

SUBSIDENCE ASSESSMENT

Susceptibility	Low
Primary Mechanism	None - consolidated terrain
Annual Rate	<1 mm/yr (est. USGS)
Historic Loss	None documented
Aquifer Compaction	None - aquifer not over-pumped
SGMA Status	Not applicable - not in SGMA basin

BENCHMARK IMPLICATIONS

Monument Risk	Low - monuments stable
Elevation Validity	Good - no active subsidence
GNSS Recommendation	Standard benchmark verification adequate

SURVEY ACTION

Required Action	No special action required
Dataset	CA DWR SGMA Bulletin 118 2024.1
Coverage	California critically overdrafted basins only - full US coverage pending

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)

Site Class	D (default)
Risk Category	II

WIND / SNOW / ICE / FROST (ASCE 7-22)

Basic Wind Speed	92 mph
Ground Snow Load	5.8 psf
Winter Wind (W2)	0.35
Ice Thickness	None
Frost Depth	12 in minimum

Est. from IECC Zone 3C — AFI data unavailable

CLIMATE ZONE (IECC 2021): Zone 3C · Marine · Marine - mild year-round, coastal influence (SF, LA coast) County: Napa

AI CLIMATE SUMMARY — RESIDENTIAL DESIGN

This marine climate features mild temperatures year-round with moderate winds up to 92 mph, so your building envelope should prioritize air sealing and impact-resistant glazing rather than heavy insulation—think continuous weather barriers and well-sealed windows. Orient primary glazing toward the south and east to capture available winter sun, but shade west-facing glass aggressively with overhangs, vegetation, or external louvers to prevent afternoon overheating and glare. The key passive move here is to design deep roof overhangs (3–4 feet minimum) that provide year-round sun control while allowing the ocean breeze to naturally ventilate the building, reducing your reliance on mechanical cooling during the warmer months.

AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU

This coastal marine site presents moderate wind exposure and light snow loading, with well-drained gravelly soils anticipated to support conventional shallow foundations. Seismic parameters are not currently available in the database; a geotechnical investigation may warrant confirmation of site class and soil bearing capacity. Frost depth of approximately 12 inches should be verified with the local building official.

FOUNDATION

Type Conventional spread footing anticipated

Basis Well-drained Coombs gravelly loam with good drainage characteristics suggests competent bearing soil; consult a licensed engineer to confirm bearing capacity and settlement after site investigation

STRUCTURAL IMPLICATIONS

Seismic Seismic design category not available — geotechnical investigation recommended to establish site class and confirm whether ordinary or special seismic detailing may be warranted

Wind Exposure Exposure C (open terrain) — consistent with ASCE 7-22 Risk Category II and 92 mph 3-second gust wind speed

Wind Standard wind load design anticipated for Exposure C and Risk Category II; no hurricane exposure suggested — verify exposure category with site reconnaissance

Snow Light snow load of 5.8 psf anticipated — standard roof design likely adequate; winter wind factor W2=0.35 may warrant verification of drift and exposure conditions

INVESTIGATIONS & INSPECTIONS

Geotech Required Yes — investigation recommended to confirm site class, bearing capacity, settlement potential, and seismic parameters; frost depth verification with local authority also suggested

Liquefaction Screen Screening recommended — Site Class D and well-drained soils suggest low liquefaction risk, but confirmation through site-specific investigation advised

Special Inspections Not anticipated at this planning stage; requirements will depend on seismic design category determination after geotechnical investigation

Code Edition ASCE 7-22 / IBC 2021 (planning reference only)

Site Class Note Site Class D assumed for planning purposes — licensed geotechnical engineer investigation strongly recommended to confirm soil properties, bearing capacity, and seismic classification

SOILS — USDA SOIL SURVEY

MAP UNIT: Coombs gravelly loam, 2 to 5 percent slopes

SERIES: Coombs

DRAINAGE CLASS: Well drained

HYDRO GROUP: C

SOIL ORDER: Alfisols

SUBORDER: Xeralfs

AI SOIL INTELLIGENCE — CLAUDE HAIKU

Coombs gravelly loam is an Alfisol (Xeralf) that formed in alluvial and colluvial materials derived from mixed rock sources in the California Coast Ranges and inland valleys. This well-drained soil occupies gentle slopes (2–5%) and is characterized by a gravelly loam surface layer overlying a clay loam substratum. Walking onto the site, expect moderate gravel content in the upper profile, firm to very firm subsoil conditions, and good trafficability except during wet periods when clay-rich layers may become plastic.

Formed in mixed alluvial and colluvial deposits; footslope to terrace positions; annual grassland and oak woodland vegetation.

DRAINAGE

Class Well drained

Seasonal Saturation None within 60 inches

Flooding Risk Minimal on 2–5% slopes

Ponding Risk None

ENGINEERING

Bearing Capacity 3,500–4,500 psf (shallow foundations)

Rating Good

Shrink-Swell Moderate — clay loam substratum expands and contracts seasonally

Frost Action Low

Cut Suitability Good (gravel-rich surface and subgrade material)

Fill Suitability Fair — clay loam substratum suitable only when moisture-conditioned; compact to 90% Standard Proctor

SEPTIC

Suitability Conditional

Notes Perc test required — clay loam substratum typically exhibits slow to moderate permeability; placement depth and drain field sizing must account for restrictive layer

HAZARDS

Excavation Moderate — 20–35% gravel content; requires heavy equipment; boulder-sized fragments possible in lower profile

Expansive Clay Low to moderate — clay substratum shows mild to moderate expansion potential

Erosion Low to moderate on 2–5% slopes; rilling and gully risk in disturbed or denuded areas

Other None anticipated

 OIL & GAS WELLS

California — within 10mi radius

8

total

0

active

7

plugged

By status:

7

Plugged

1

Idle

By type: 8 DH

Lease / Well	Status	Type	Operator	Depth	Dist
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
McKenzie-Hazard Syndicate	Plugged	DH	McKenzie-Hazard Syndicate	—	7.4 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.6 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.8 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	9.5 mi
Capell Oil Corp.	Idle	DH	Capell Oil Corp.	—	9.7 mi
Cordelia Oil & Gas Co.	Plugged	DH	Cordelia Oil & Gas Co.	—	12 mi

AI Assessment — Well Proximity

Well proximity is not a planning concern for this site. The nearest well, a plugged Mobil Oil dry hole 7.2 miles away, is well beyond any actionable distance threshold. No active producers, injection wells, or other well types exist within relevant review distances.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

200
total in search area

10
within 5 mi

108
within 25 mi

Site Name	Status	Commodities	Distance
Roderick Sand and Gravel Pit	Occurrence	Sand/Gravel	1.6 mi
Basalt Rock Co	Producer	Clay	2 mi
Cicero Pumice Quarry	Occurrence	PUM	2.9 mi
Mount George King Pumice Deposit.	Occurrence	PUM	2.9 mi
Basalt Rock Co.	Occurrence	DIT	3.1 mi
Mt George Pumice Deposit	Producer	PUM	3.1 mi
Wilson Pumice Quarry	Occurrence	PUM	3.7 mi
Unknown	Occurrence	Mercury	4.4 mi
Silva Pumice Quarry	Occurrence	PUM	4.7 mi
Silva Pumice Quarry	Producer	PUM	5 mi
Walker Pumice Quarry	Occurrence	PUM	5.1 mi
Walker Pumice Quarry	Producer	PUM	5.1 mi
Mountain	Occurrence	Mercury	5.4 mi
Unknown	Occurrence	Mercury	7.6 mi
Unknown	Occurrence	Stone/Crushed	7.8 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Prospect	Occurrence	Chromium	8.8 mi
Unnamed Location	Occurrence	Chromium	9 mi
Unknown	Occurrence	Chromium	9.1 mi
Bella Oak	Past Producer	Mercury	9.4 mi
Unknown	Occurrence	MG	9.7 mi
Serres Pit	Past Producer	Sand/Gravel	10.4 mi
Unnamed Location	Unknown	Sand/Gravel	10.5 mi
Conn Valley	Occurrence	Manganese	10.6 mi

Showing 25 of 200 deposits. Full dataset via USGS MRDS.

AI Assessment — Mining Context

Three active mineral producers operate within 5 miles of the site. Basalt Rock Co (2 miles, clay) and Mt George Pumice Deposit (3.1 miles, pumice) and Silva Pumice Quarry (5 miles, pumice) should be anticipated to generate heavy truck traffic on access roads during operating hours, dust emissions, and potential noise from blasting or crushing operations. No inactive deposits warrant Phase 1 assessment consideration based on proximity.

Source: USGS Mineral Resources Data System (MRDS). Status: Producer = active/recent; Past Producer = historical; Prospect = explored not mined; Occurrence = mineralization noted.

ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

6
Endangered

6
Threatened

12
Total w/ Critical Habitat

6 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

COMMON NAME	SCIENTIFIC NAME	STATUS
California tiger Salamander	<i>Ambystoma californiense</i>	Endangered
Conservancy fairy shrimp	<i>Branchinecta conservatio</i>	Endangered
Contra Costa goldfields	<i>Lasthenia conjugens</i>	Endangered
Soft bird's-beak	<i>Cordylanthus mollis</i> ssp. <i>mollis</i>	Endangered
Suisun thistle	<i>Cirsium hydrophilum</i> var. <i>hydrophilum</i>	Endangered
Vernal pool tadpole shrimp	<i>Lepidurus packardii</i>	Endangered
Alameda whipsnake (=striped racer)	<i>Masticophis lateralis euryxanthus</i>	Threatened
California red-legged frog	<i>Rana draytonii</i>	Threatened
Delta smelt	<i>Hypomesus transpacificus</i>	Threatened
Northern spotted owl	<i>Strix occidentalis caurina</i>	Threatened
Vernal pool fairy shrimp	<i>Branchinecta lynchi</i>	Threatened
western snowy plover	<i>Charadrius nivosus nivosus</i>	Threatened

AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Section 7 Consultation Trigger Analysis **38.3459, -122.3047 – Solano County Region** Section 7 consultation with USFWS will be triggered if the proposed project involves federal nexus (permits, funding, or approvals) and may affect any of the 12 listed species or their critical habitat. The most planning-relevant species for terrestrial development include California tiger salamander and vernal pool invertebrates (fairy shrimp, tadpole shrimp), which indicate seasonal wetland constraints, and California red-legged frog, which suggests riparian/aquatic buffers; aquatic projects must also evaluate Delta smelt and Suisun thistle impacts. The recommended next step is to conduct a formal biological assessment identifying site-specific habitat suitability, seasonal presence/absence surveys for vernal pool and wetland-dependent species, and delineation of critical habitat overlays to determine project footprint feasibility and required mitigation or avoidance measures. Early coordination with USFWS and CDFW is essential to establish project-specific conservation requirements before design finalization. **For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.**

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

NATIVE PLANTS - iNaturalist Research-Grade within 10mi

57	5051	2026-07-04
Unique Species	Total Observations	Most Recent
COMMON NAME	SCIENTIFIC NAME	LAST OBS
American black nightshade	<i>Solanum americanum</i>	2026-05-29
Bearded Clover	<i>Trifolium barbigerum</i>	2026-05-15
bird's foot cliffbrake	<i>Pellaea mucronata</i>	2026-04-28
Blow Wives	<i>Achyrachaena mollis</i>	2026-05-20
blue oak	<i>Quercus douglasii</i>	2026-05-09
Bolander's Phacelia	<i>Phacelia bolanderi</i>	2026-04-27
box elder	<i>Acer negundo</i>	2026-05-18
Braun's Giant Horsetail	<i>Equisetum telmateia braunii</i>	2026-06-22
California bay	<i>Umbellularia californica</i>	2026-06-24
California beeplant	<i>Scrophularia californica</i>	2026-05-20
California black oak	<i>Quercus kelloggii</i>	2026-05-15
California buckeye	<i>Aesculus californica</i>	2026-06-02
California Dutchman's Pipe	<i>Aristolochia californica</i>	2026-05-26
California milkwort	<i>Rhinotropis californica</i>	2026-05-15
California mugwort	<i>Artemisia douglasiana</i>	2026-06-02
California poppy	<i>Eschscholzia californica</i>	2026-06-24
California Wild Rose	<i>Rosa californica</i>	2026-05-09
Canyon liveforever	<i>Dudleya cymosa</i>	2026-05-30
-	<i>Castilleja densiflora densiflora</i>	2026-05-09
clay mariposa lily	<i>Calochortus argillosus</i>	2026-05-15
coast live oak	<i>Quercus agrifolia</i>	2026-05-13
coastal woodfern	<i>Dryopteris arguta</i>	2026-05-20
common selfheal	<i>Prunella vulgaris</i>	2026-06-22
Diogenes' lantern	<i>Calochortus amabilis</i>	2026-05-15
distant phacelia	<i>Phacelia distans</i>	2026-05-15
-	<i>Downingia concolor concolor</i>	2026-05-15
Dwarf Brodiaea	<i>Brodiaea terrestris</i>	2026-05-15
Elegant Clarkia	<i>Clarkia unguiculata</i>	2026-05-16
goldback fern	<i>Pentagramma triangularis</i>	2026-05-20

COMMON NAME	SCIENTIFIC NAME	LAST OBS
Ithuriel's Spear	<i>Triteleia laxa</i>	2026-06-15
maroonspot calicoflower	<i>Downingia concolor</i>	2026-05-15
narrowleaf milkweed	<i>Asclepias fascicularis</i>	2026-06-25
narrowleaf mule-ears	<i>Wyethia angustifolia</i>	2026-05-15
Narrowleaf Willow	<i>Salix exigua</i>	2026-06-24
needleleaf navarretia	<i>Navarretia intertexta</i>	2026-05-15
northern California black walnut	<i>Juglans hindsii</i>	2026-06-24
orange bush monkeyflower	<i>Diplacus aurantiacus</i>	2026-05-29
Oregon Ash	<i>Fraxinus latifolia</i>	2026-06-02
Pacific Bleeding Heart	<i>Dicentra formosa</i>	2026-05-04
Pacific madrone	<i>Arbutus menziesii</i>	2026-07-04
Pacific poison oak	<i>Toxicodendron diversilobum</i>	2026-06-02
pineapple-weed	<i>Matricaria discoidea</i>	2026-05-20
Redwood Sorrel	<i>Oxalis smalliana</i>	2026-05-28
small-headed clover	<i>Trifolium microcephalum</i>	2026-05-15
Superb Mariposa Lily	<i>Calochortus superbus</i>	2026-06-11
thimble clover	<i>Trifolium microdon</i>	2026-05-15
tomcat clover	<i>Trifolium willdenovii</i>	2026-05-15
Toyon	<i>Heteromeles arbutifolia</i>	2026-06-19
trailing blackberry	<i>Rubus ursinus</i>	2026-06-24
Tree Clover	<i>Trifolium ciliolatum</i>	2026-05-15
valley oak	<i>Quercus lobata</i>	2026-06-24
wavy-leafed soap plant	<i>Chlorogalum pomeridianum</i>	2026-05-20
western sycamore	<i>Platanus racemosa</i>	2026-05-17
white brodiaea	<i>Triteleia hyacinthina</i>	2026-05-15
white-tipped clover	<i>Trifolium variegatum</i>	2026-05-15
Winecup Clarkia	<i>Clarkia purpurea</i>	2026-05-15
Yellow Mariposa Lily	<i>Calochortus luteus</i>	2026-06-11

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Ecological Assessment: 38.3459, -122.3047 This site supports a ****mixed oak woodland and chaparral community****, characteristic of the inner North Coast Ranges. The dominant species—blue oak, California black oak, California bay, and California buckeye—indicate well-drained upland slopes with seasonal moisture, while presence of Braun's Giant Horsetail and clay mariposa lily suggests areas with intermittent seepage or springline habitat. Grading activities should anticipate ****variable soil drainage across the site****, with clay-rich soils in some areas retaining winter moisture but drying significantly by summer; preservation of drainage patterns and riparian corridors will be critical to maintaining native plant communities. The diverse wildflower component (California poppy, mariposa lily, phacelia species) indicates this ecosystem depends on natural seasonal flooding and infiltration cycles that any earthwork should carefully respect. ****Consult a licensed botanist for site-specific vegetation assessment.****

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

635	154	91	1439
Venomous Snakes	Venomous Spiders	Stinging Insects	Toxic Plants
VENOMOUS SNAKES			
SPECIES		OBSERVATIONS	
Crotalus oreganus oreganus		635 obs	
VENOMOUS SPIDERS			
SPECIES		OBSERVATIONS	
Latrodectus hesperus		154 obs	
STINGING INSECTS			
SPECIES		OBSERVATIONS	
Dasymutilla aureola		91 obs	
Dasymutilla californica		91 obs	
Dasymutilla sackenii		91 obs	
TOXIC PLANTS			
SPECIES		OBSERVATIONS	
Toxicodendron diversilobum		1100 obs	

SPECIES	OBSERVATIONS
<i>Datura wrightii</i>	57 obs
<i>Datura stramonium</i>	57 obs
<i>Datura ferox</i>	57 obs
<i>Conium maculatum</i>	199 obs
<i>Urtica gracilis</i>	83 obs
<i>Urtica gracilis holosericea</i>	83 obs

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU

Field Safety Brief – Northern California Survey Area ****Highest-Priority Hazards:**** Poison oak (*Toxicodendron diversilobum*) dominates this region with 1,439 observations and poses the greatest risk to survey crews—it's widespread, easily contacted, and causes severe dermatitis in most people. Northern Pacific rattlesnakes (*Crotalus oreganus oreganus*) are the second concern with 635 observations; while bites are rare, they're medically serious. Black widow spiders and various stinging insects are lower-frequency but still present threats. ****Practical Avoidance:**** Wear long sleeves, long pants, and closed-toe boots; learn to identify poison oak by its three-leaflet pattern and avoid contact. Watch for rattlesnakes in rocky areas and tall grass—make noise while moving, watch foot placement, and never reach into areas you can't see. Inspect your body and clothing daily for ticks, and wash exposed skin thoroughly at day's end. ****Seasonal Note:**** Late spring through early fall brings peak snake and insect activity; poison oak irritant oils are potent year-round but peak in warmer months. Early morning surveys reduce snake encounters. Always verify current conditions locally before fieldwork.

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FCC MICROWAVE PATHS — Licensed beam paths crossing within 1320ft of site

3	545ft	3
Beam Paths	Closest Beam	Fixed P2P

TX CALL	RX CALL	TYPE	PATH MI	CLOSEST
WBO61	WBO59	Fixed Point-to-Point	—	545ft
KYH37	WQAT627	Fixed Point-to-Point	—	927ft
WRCR329	WQAT627	Fixed Point-to-Point	—	1289ft

Source: FCC ULS microwave license database — PostGIS spatial query — Paths within 1320ft (0.25mi) of site — Towers may be 50+ miles away — FCC Part 101 licensed paths

FAA AIRWAYS — ATS Routes crossing site area

26	13	2	11
Total Airways	Jet Routes	Victor Airways	Other Routes

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
J1	Jet Route	High	18,000 ft	—	303.8°	121.4°
J110	Jet Route	High	18,000 ft	—	137.9°	318.2°
J126	Jet Route	High	18,000 ft	—	325.5°	144.0°
J143	Jet Route	High	18,000 ft	—	325.0°	145.8°
J3	Jet Route	High	18,000 ft	—	342.8°	161.8°
J32	Jet Route	High	18,000 ft	—	19.2°	199.6°
J501	Jet Route	High	18,000 ft	—	317.1°	135.4°
J58	Jet Route	High	18,000 ft	—	65.3°	245.9°
J65	Jet Route	High	18,000 ft	—	304.1°	121.1°
J80	Jet Route	High	18,000 ft	—	65.3°	245.9°
J84	Jet Route	High	18,000 ft	—	50.8°	231.0°
J88	Jet Route	High	18,000 ft	—	308.0°	127.3°
J94	Jet Route	High	18,000 ft	—	65.3°	245.9°
Q1	Q-Route	High	—	—	334.5°	154.1°
Q120	Q-Route	High	—	—	38.1°	219.8°
Q122	Q-Route	High	—	—	37.8°	219.6°
Q124	Q-Route	High	—	—	37.8°	219.6°
Q138	Q-Route	High	—	—	47.7°	229.0°
Q3	Q-Route	High	—	—	164.0°	344.6°
Q5	Q-Route	High	—	—	161.4°	341.7°
T257	T-Route	Low	—	—	321.1°	140.5°
T259	T-Route	Low	—	—	41.6°	221.8°
T261	T-Route	Low	—	—	31.4°	211.7°
T263	T-Route	Low	—	—	308.8°	128.2°
V107	Victor Airway	Low	7,000 ft	—	295.4°	114.2°

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
V108	Victor Airway	Low	4,500 ft	—	117.7°	296.8°

Source: FAA ATS Route FeatureServer — Airways shown cross the site bounding area and may not pass directly overhead — MEA = Minimum En Route Altitude

RAILROAD PROXIMITY — FRA North American Rail Network

8
Rail Lines

OWNER / OPERATOR	TYPE	TRACKS	PASSENGER	STRACNET	SUBDIVISION
CFNR	Other	1	No	No	—
CFNR	Yard	1	No	No	—
CFNR	Mainline	1	No	No	VALLEJO
SMRT	Mainline	1	No	No	BRAZOS JCT.
NVRR	Mainline	1	No	No	MARTINEZ
USG	Other	1	No	No	—
CFNR	S	1	No	No	—
NVRR	Yard	1	No	No	—

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

200 91 66 5 14 6
Total Observations Total Species Birds Mammals Reptiles Amphibians

BIRDS (66 species)

COMMON NAME	Scientific Name	OBS
Wild Turkey	<i>Meleagris gallopavo</i>	9
Snowy Egret	<i>Egretta thula</i>	7
California Quail	<i>Callipepla californica</i>	5
Western Bluebird	<i>Sialia mexicana</i>	5
California Towhee	<i>Melozone crissalis</i>	5
Great Egret	<i>Ardea alba</i>	4
Red-tailed Hawk	<i>Buteo jamaicensis</i>	4
Black Phoebe	<i>Sayornis nigricans</i>	4
Western Screech-Owl	<i>Megascops kennicottii</i>	4
Cooper's Hawk	<i>Astur cooperii</i>	3
Anna's Hummingbird	<i>Calypte anna</i>	3
Ash-throated Flycatcher	<i>Myiarchus cinerascens</i>	3
Turkey Vulture	<i>Cathartes aura</i>	3
Mallard	<i>Anas platyrhynchos</i>	3
Bushtit	<i>Psaltiriparus minimus</i>	3

+ 51 additional species

MAMMALS (5 species)

COMMON NAME	Scientific Name	OBS
Black-tailed Jackrabbit	<i>Lepus californicus</i>	4
Coyote	<i>Canis latrans</i>	2
Mule Deer	<i>Odocoileus hemionus</i>	2
Eastern Fox Squirrel	<i>Sciurus niger</i>	1
Striped Skunk	<i>Mephitis mephitis</i>	1

REPTILES (14 species)

COMMON NAME	Scientific Name	OBS
Western Fence Lizard	<i>Sceloporus occidentalis</i>	9
Pacific Gopher Snake	<i>Pituophis catenifer catenifer</i>	8
Northern Pacific Rattlesnake	<i>Crotalus oreganus oreganus</i>	6
California King Snake	<i>Lampropeltis californiae</i>	3
Pond Slider	<i>Trachemys scripta</i>	2
Western Skink	<i>Plestiodon skiltonianus</i>	2

COMMON NAME	Scientific Name	OBS
Ring-necked Snake	<i>Diadophis punctatus</i>	1
Common Garter Snake	<i>Thamnophis sirtalis</i>	1
Western Yellow-bellied Racer	<i>Coluber constrictor mormon</i>	1
Pacific Ringneck Snake	<i>Diadophis punctatus amabilis</i>	1
Northwestern Pond Turtle	<i>Actinemys marmorata</i>	1
Common Sharp-tailed Snake	<i>Contia tenuis</i>	1
North American Racer	<i>Coluber constrictor</i>	1
Southern Alligator Lizard	<i>Elgaria multicarinata</i>	1

AMPHIBIANS (6 species)

COMMON NAME	Scientific Name	OBS
Pacific chorus frog	<i>Pseudacris regilla</i>	5
American Bullfrog	<i>Lithobates catesbeianus</i>	3
Foothill Yellow-legged Frog	<i>Rana boylei</i>	3
Rough-skinned Newt	<i>Taricha granulosa</i>	1
Western Toad	<i>Anaxyrus boreas</i>	1
California Giant Salamander	<i>Dicamptodon ensatus</i>	1

AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU

Ecological Character Assessment This site in Sonoma County, California supports a diverse wildlife community indicative of oak woodland, grassland, and riparian transition zones with proximity to freshwater resources. The presence of sensitive species including Foothill Yellow-legged Frog (California Species of Special Concern), Northwestern Pond Turtle (California Species of Special Concern), and California Giant Salamander indicates aquatic or seasonally wet habitat features that warrant protective consideration, along with adequate structural complexity in upland areas. The mammal and bird assemblage—dominated by generalist species such as coyote, wild turkey, and common raptors—suggests moderate disturbance tolerance, though the diversity of native reptiles and amphibians reflects relatively intact riparian corridors and seasonal water availability essential for breeding. Site conditions appear to support mixed-use habitat with functional connectivity to surrounding natural areas, though development pressures typical of the region should be evaluated in the context of California Department of Fish and Wildlife regulations governing state-listed and sensitive species.
For planning purposes only. Consult a qualified biologist for site-specific biological assessments.

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

✳️ PASSIVE + ACTIVE SOLAR PLANNING — Lat 38.35°

1 • AVERAGE SUN HOURS

Period	Peak Hrs/Day	Notes
Annual average	6.6 hrs	Overall solar resource
Summer average	9 hrs	June–August
Winter average	3.9 hrs	December–February
Worst month (Dec)	3.2 hrs	Size battery storage to this

2 • IDEAL SOLAR ARRAY PITCH

Optimal tilt = 38.3° (latitude) → 9/12 pitch

Pitch	Angle	% of Optimal	Notes
4/12	18.4°	97%	good
5/12	22.6°	97.6%	good
6/12	26.6°	98.2%	↪ excellent
7/12	30.3°	98.8%	↪ excellent
8/12	33.7°	99.3%	↪ excellent
9/12	36.9°	99.8%	↪ annual optimal
10/12	39.8°	99.8%	↪ excellent
12/12	45°	99%	↪ excellent

Going from 5/12 to 6/12 at framing costs ~\$800–1,200 one time. On a 10kW system that’s +723 kWh/yr (~\$108 value/yr). Over 25yr panel life: ~\$2710 total gain.

3 • SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

Surface	Winter	Summer	Design Note
South glass (SHGC 0.4)	51	31	Minimize south glass
South glass (SHGC 0.6)	76	46	Clear/uncoated glass
North glass (any)	10	13	Negligible — diffuse only
East / West glass	27	104	⚠ Minimize — afternoon load
Roof — dark (absorb 0.95)	236	301	Highest load surface
Roof — light (absorb 0.35)	87	111	Cool roof — significant reduction
Roof — metal (absorb 0.25)	62	79	Best roof option for heat
South wall — dark	94	69	
South wall — light	31	23	Light color = major savings
Hardscape — asphalt	181	301	⚠ Heat island — avoid near structure
Hardscape — concrete	124	206	Also reflects onto adjacent walls
Hardscape — light pave	67	111	Recommended within 50ft
Pool surface	23	38	Low absorb + evaporative cooling

Concrete reflection onto south wall: ~4 BTU/hr-ft² additional load per 100sf of adjacent hardscape.

Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 · ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

Pitch	Dark Roof	Light Roof	Reduction vs Flat
0/12	301 BTU	111 BTU	baseline (worst)
4/12	241 BTU	89 BTU	-20% dark / -20% light
6/12	196 BTU	72 BTU	-35% dark / -35% light
8/12	166 BTU	61 BTU	-45% dark / -45% light
12/12	135 BTU	50 BTU	-55% dark / -55% light

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 · THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

Assembly	Lag	Peak Interior Load	Note
2x4 frame (R-15)	0.6 hrs	~15:36 PM	R-value only — no mass benefit
2x6 frame (R-23)	0.9 hrs	~15:54 PM	R-value only — no mass benefit
2x8 frame (R-30)	1.2 hrs	~16:12 PM	R-value only — no mass benefit
2x10 frame (R-37)	1.5 hrs	~16:30 PM	good
2x12 frame (R-44)	1.8 hrs	~16:48 PM	good
ICF 6" core	7 hrs	~22:00 PM	✓ shifts past sunset — natural vent window
8" CMU block	6 hrs	~21:00 PM	✓ shifts past sunset — natural vent window
12" concrete	8.5 hrs	~23:30 PM	✓ shifts past sunset — natural vent window

In 3C Marine: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 · TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

Tree Ht	Shadow Reaches	Summer hrs/day	Equinox hrs/day	Winter hrs/day	WUI note
10ft	All year	1.1	1	1.2	OK
20ft	All year	2.1	1.9	2.4	OK
30ft	All year	3.1	2.9	3.7	Zone 2 check
40ft	All year	4	3.8	5.2	△ verify
50ft	All year	4.8	4.6	7.3	△ verify
60ft	All year	5.6	5.5	9.3	✗ Zone 1

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.

Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.

△ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 · ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

Standard	Unbalanced (1/150)	Balanced† (1/300)	Notes
Code minimum	8 sq in	4 sq in	Moisture protection only
Thermal comfort	24 sq in	12 sq in	Keeps attic within 20°F of outdoor
WUI mesh 1/16"	60 sq in	30 sq in	2.5× compensation — mesh clogs
WUI intumescent	24 sq in	12 sq in	Self-closing — no compensation needed

Pitch factor: 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45

Color factor: Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45

Scale: multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor

†**Balanced** = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)

WUI conflict: Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.

△ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

🔍 AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Your Home's Solar Potential and Energy Strategy Your location gets about 6.6 hours of useful sunlight per day on average—think of it like having a solar battery that charges reliably but isn't at full strength in winter. Your roof's slope of about 9 inches per 12 inches is nearly perfect for solar panels, so you've got good bones for that investment. Here's where it gets interesting: a dark roof in summer acts like a giant heat magnet, soaking up roughly three times more energy than a light-colored one, which means you're either cooling harder or choosing a lighter shade to fight that. West-facing windows are your real summer enemy—they let in afternoon heat when your air conditioner is already tired—so consider shading them with overhangs or plants, because that afternoon sun exposure is nearly double what hits your south windows. A leafy tree on your south side is a smart move: it'll shade you in summer when you don't want that heat, then drop its leaves in winter to let warming sun through when you do. Thermal mass is just a fancy term for anything dense in your home (concrete, brick, stone) that soaks up heat during the day and slowly releases it at night, giving you natural temperature smoothing. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.



Technical Brief — for architects, engineers, and surveyors

SITE INTELLIGENCE BRIEFING **Napa County, CA | 38.34586°N, 122.30466°W**

Structural Design Drivers

IECC Zone 3C (Marine) governs this site. With an annual freezing index of 0°F-days and no snow load data provided, cold-climate load cases are not a structural driver. Wind design requirements for Zone 3C apply but specific gust speeds are not included in this dataset; standard ASCE 7 procedures for the California coast will apply. Seismic design parameters are not available in the provided data, creating a critical gap: Napa County is within an active seismic region (proximity to San Andreas and Hayward faults is likely but not confirmed by this search). Do not proceed with structural system selection until seismic design category and spectral response parameters are obtained from USGS hazard maps or a site-specific seismic study.

Geotechnical Considerations

The site is underlain by Coombs gravelly loam, 2–5 percent slopes, classified in NRCS Hydrologic Group C with well-drained characteristics. This soil type supports shallow foundations under typical conditions; group C classification indicates moderate infiltration capacity with lower permeability than groups A or B. The well-drained designation suggests subsurface water is not a primary concern for bearing capacity, but the gravelly loam texture (confirmed) paired with slopes of 2–5 percent warrants a subsurface investigation to confirm depth to bedrock and lateral variability across the building footprint. Frost depth is not provided; obtain California Building Standards frost depth for Napa County (likely 12–18 inches) before setting footing depths.

Hazard Assessment

No Quaternary faults appear within the search radius, but this absence does not confirm seismic safety; a fault-rupture hazard assessment aligned with CGS Alquist-Priolo guidelines is mandatory for any structure in Napa County. No subsidence basin data is recorded, removing subsidence as a flag. Wind and snow hazards are not quantified in this dataset. Wildfire risk is not included; cross-reference Cal Fire hazard maps and defensible-space requirements for this latitude-longitude pair.

Site Context & Mining Activity

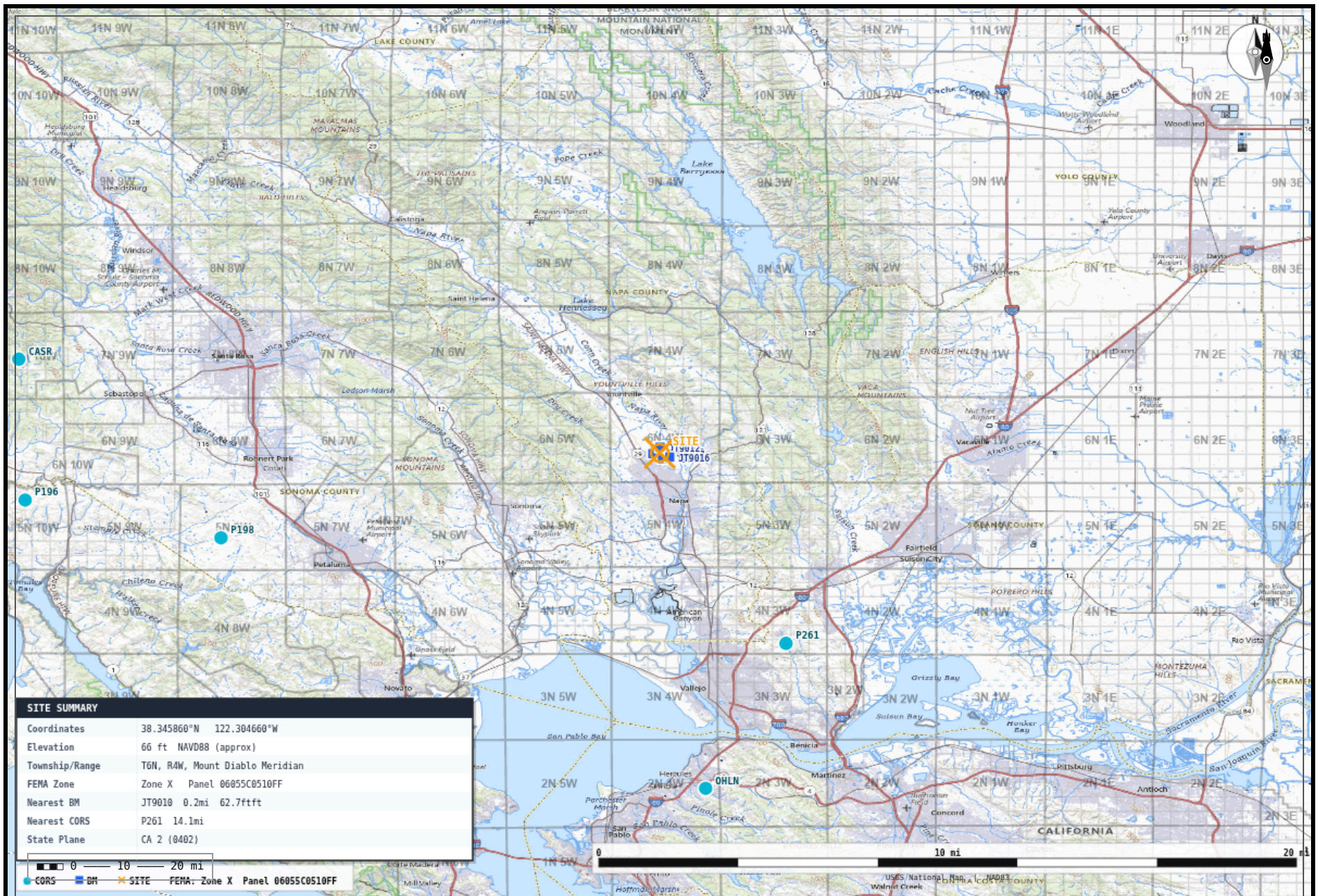
Three active aggregate and pumice operations exist within 5 miles: Basalt Rock Co (2 miles, clay), Mt George Pumice (3.1 miles), and Silva Pumice Quarry (5 miles). While none pose direct ground hazard, vibration from quarry blasting should be assessed if sensitive equipment or structures are planned; obtain blast records from operators. Eight oil/gas wells exist within 10 miles; all are either plugged or non-producing, eliminating subsurface fluid withdrawal risk. Climate Zone 3C (marine) implies modest heating/cooling loads and corrosion risk typical of coastal California; specify corrosion-resistant materials for exposed metals.

****Bottom line:**** Obtain seismic design parameters and confirm frost depth before foundation design; conduct Phase I subsurface investigation to verify bearing capacity and verify depth to bedrock given slopes and gravelly loam composition.



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Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.

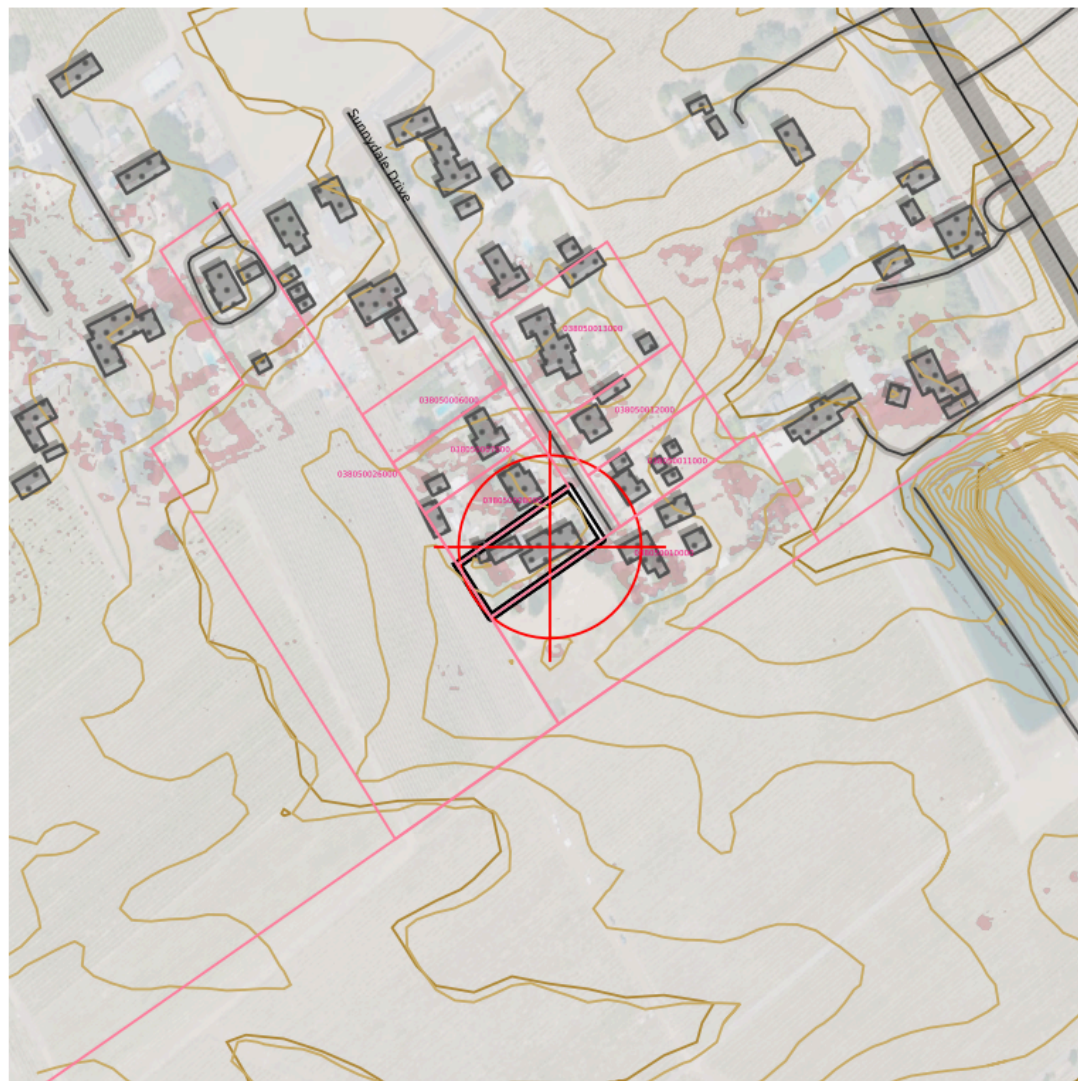


TOPOSCOUT
FIELD MAP

TopoScout Field Map
Lat: 38.34586 Lng: -122.30466
38.345860°N 122.304660°W | Radius: 30 mi

Generated: July 12, 2026
Base: USGS National Map
Generated by TopoScout - outkastdesigns.com

TOPOSCOUT SITE PLAN PREVIEW | \$19 UPGRADE FOR AUTOCAD DXF FILE



INCLUDED LAYERS



0 — 400 ft State Plane · NAD83 · US Feet

- Contours (Index/Major/Minor/1ft)
- PLSS Section/Township/QQ
- Roads Major/Local/Trail + ROW
- Buildings
- Hydro
- Power Lines
- Pipeline
- Railroad
- Benchmarks + Labels
- County Boundary
- OTLS Grid
- Site Info
- Elevation Labels
- 3D Polylines (all geometry)

✂ FIELD RESOURCE LINKS

PRE-MISSION PLANNING

Trimble GNSS Planning Online –

<https://trimblegnssplanningonline.com/>

Satellite visibility, DOP charts, sky plots

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Drone airspace authorization, TFRs, restricted areas

Aloft (Kittyhawk) – <https://www.aloft.ai/>

Field-friendly drone airspace map and LAANC authorization

NOAA Space Weather — Live Kp –

<https://www.swpc.noaa.gov/products/planetary-k-index>

Real-time Kp index, geomagnetic storm alerts

POST-PROCESSING

NGS OPUS – <https://geodesy.noaa.gov/OPUS/index.jsp>

Online Positioning User Service – static GNSS processing

NGS OPUS-RS – <https://geodesy.noaa.gov/OPUSI/index.jsp>

Rapid static processing – sessions as short as 15 minutes

CORS Data Download – <https://geodesy.noaa.gov/CORS/cors-data.shtml>

Raw RINEX observation files from CORS stations

FEMA & FLOOD

FEMA Flood Map Service Center –

<https://msc.fema.gov/portal/home>

FIRM panels, flood zone lookup by address or coordinates

FEMA Elevation Certificate – [https://www.fema.gov/flood-](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

[insurance/work-with-nfip/elevation-certificate](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

Current EC form, instructions, LOMA/LOMR submission guidance

BASE STATION & NETWORK

NOAA CORS Network – <https://geodesy.noaa.gov/CORS/cors-data.shtml>

Continuously operating reference stations, data download

NGS CORS Station Map –

<https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml>

Interactive map of all CORS stations with status

IGS Station Monitor – <https://www.igs.org/network>

International GNSS Service – global base station health

REFERENCE & CALCULATIONS

NGS Datasheets – <https://www.ngs.noaa.gov/cgi-bin/datasheet.prl>

Benchmarks, control points, passive monuments

NGS Coordinate Conversion (NCAT) –

<https://geodesy.noaa.gov/NCAT/>

NADCON5, datum transforms, state plane conversions

NOAA Magnetic Declination –

<https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml>

Declination calculator for any location and date

NGS Geoid Models – [https://geodesy.noaa.gov/GEOID/](https://geodesy.noaa.gov/GEOID/GEOID18)

GEOID18 and current model downloads, GEOID calculator

DRONE OPERATIONS

FAA Part 107 — Commercial Rules –

https://www.faa.gov/uas/commercial_operators

Full Part 107 regulatory summary for commercial UAS operations

FAA DroneZone – <https://faadronezone.faa.gov/>

Drone registration, operator certificates, waiver management

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Pre-flight airspace check, TFRs, LAANC authorization

FAA Part 107 Waivers –

https://www.faa.gov/uas/commercial_operators/part_107_waivers

Waiver applications for operations beyond standard 107 rules

Aloft — Field Airspace Map – <https://www.aloft.ai/>

Mobile-friendly real-time airspace, LAANC, flight logging

AIR QUALITY & WILDFIRE SMOKE

[AirNow Fire & Smoke Map](https://www.fire.airnow.gov)

<https://www.fire.airnow.gov>

Current fire locations, smoke plumes, PM2.5 readings by location

[AirNow Interactive Map](https://gispub.epa.gov/airnow)

<https://gispub.epa.gov/airnow>

Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

[NOAA Smoke Forecast](https://airquality.weather.gov)

<https://airquality.weather.gov>

48-hr smoke concentration forecast, HRRR-Smoke model

